Pontsho Mbizo

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Summary

Final-year Electrical & Computer Engineering student (UCT, ECSA-accredited) with four industry placements across defence, mining and infrastructure. I build hardware and embedded systems that work, and I check them against measurements: a gantry robot with closed-loop control and on-device vision, FPGA and microcontroller simulators, and Excel tools adopted by an engineering team. Looking for a graduate hardware engineering role.

Projects

Weeding gantry robot, final-year research project (EEE4022S) *2026*

- Built a low-cost three-axis gantry (ESP32-C3, Raspberry Pi Zero W) that photographs a bed, classifies weeds on the device and sprays each one. Wrote the Pi-side software (Python) and ESP32-C3 motion firmware with homing, stall detection and a safety watchdog.
- 245 closed-loop moves ended within one sensor count (1.07 mm on X and Y); worst settling time 1.86 s against a 3 s limit; positioning within 14 mm against a 20 mm requirement. Detection scored on a spray-grid metric: F1 0.908 for segmentation against 0.654 for YOLOv8n.

fpgasim and mcusim, in-browser hardware simulators *2025 – 2026*

- **fpgasim:** write Verilog in the browser and run it through Icarus Verilog (Docker) in an ephemeral, network-less container, with a Tang Nano 9K board model and waveform view. 69 unit tests; 272–375 ms round trip.
- **mcusim:** compile bare-metal C for an STM32F103 Blue Pill with arm-none-eabi-gcc, run it in QEMU and render the GPIO trace on a board view. 68 unit tests; 357–469 ms round trip.

scope.io, interview prep for electronics engineers *2026*

- LeetCode-style platform that grades solutions on the server from CircuitJS, fpgasim and mcusim; 26 problems written, 7 live on the demo.

Work Experience

Project Engineering Intern, Tharisa Minerals *Dec 2025 – Jan 2026*

- Built a parameterised three-phase MCC sizing tool in Excel (full-load currents, voltage drop, power factor, operating cost per motor), validated analytically and against manufacturer data (SANS 1507, WEG IE3 motors).
- Percentage-deviation check surfaces specification errors before procurement; methodology report accepted by the engineering department; tool adopted by the site team as the baseline for feeder studies.

Engineering Intern, Cybicom Atlas Defence *Jun – Jul 2025*

- Designed a 200 MHz FPGA capture system (Verilog, Intel MAX10) to non-destructively read a battery-backed SRAM that presents data for about 10 ns, 70 ns after enable, with a 5 ns timer and a 5 V to 3.3 V buffer stage.
- Built a bidirectional multi-threaded UDP client/server in C (POSIX threads) with checksums and handover documentation for a motion simulator.
- Designed and 3D-printed a parametric Fusion 360 mount that secures a 4U simulation computer against inertial loads.
- Took part in an Acceptance Test Procedure for a full motion simulation system.

Network Engineer/Artisan (Vacation Work), Tharisa Minerals *Dec 2024 – Jan 2025*

- Restored LAN connectivity across plant sites and connected new office spaces; investigated Microsoft Cloud PC as a cost-reduction option.

Electrical and Instrumentation Vacation Work, Tharisa Minerals *Dec 2023 – Jan 2024*

- Assisted Electrical Artisans and Instrumentation Technicians with maintenance and inspection of plant equipment and control systems.

Education

BSc Eng Electrical and Computer Engineering, University of Cape Town, *2022 – Present*

- Final year: High Performance Embedded Systems, DSP & ML, Controls & Instrumentation, Law for Engineers. Design project: oceanographic sensor buoy (full systems engineering lifecycle).

Additional Information

- **Software & Data:** C/C++, Python (NumPy, pandas, scikit-learn), MATLAB
- **Hardware & Firmware:** Verilog HDL, FPGA, STM32, QEMU, KiCad, Intel Quartus Prime
- **Design & Tools:** Fusion 360, Onshape, Docker, Git, Linux (Fedora), LaTeX, MS Excel
- **Languages:** English (fluent), Setswana (native), isiZulu and isiXhosa (conversational)
- **Awards / Activities:** UCT Plus Silver (45 hours role-modelling leadership, residence mentoring); Dell Young Leader; Electrical Engineering peer mentor at UCT

References available upon request.